

UNIT 9: Mutualism and co-operation

Outline

Introduction

Cooperation within species

Mutualism between species

The fight to survive

- ▶ All organisms need resources

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce
 - ▶ We've all made it this far

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce
 - ▶ We've all made it this far
- ▶ Competition is the fight for resources

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce
 - ▶ We've all made it this far
- ▶ Competition is the fight for resources
- ▶ Exploitation is using other organisms as resources

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce
 - ▶ We've all made it this far
- ▶ Competition is the fight for resources
- ▶ Exploitation is using other organisms as resources
- ▶ But positive interactions have shaped the living world just as strongly as antagonistic interactions

The fight to survive

- ▶ All organisms need resources
 - ▶ To survive, grow reproduce
 - ▶ We've all made it this far
- ▶ Competition is the fight for resources
- ▶ Exploitation is using other organisms as resources
- ▶ But positive interactions have shaped the living world just as strongly as antagonistic interactions



Positive interactions

- ▶ We will refer to interactions that benefit both sides as **mutualistic**

Positive interactions

- ▶ We will refer to interactions that benefit both sides as **mutualistic**
 - ▶ People often use **mutualism** specifically to describe interactions between different species

Positive interactions

- ▶ We will refer to interactions that benefit both sides as **mutualistic**
 - ▶ People often use **mutualism** specifically to describe interactions between different species
 - ▶ **Cooperation** is a more general term

Positive interactions

- ▶ We will refer to interactions that benefit both sides as **mutualistic**
 - ▶ People often use **mutualism** specifically to describe interactions between different species
 - ▶ **Cooperation** is a more general term

The influence of mutualism



The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation



The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ *

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ *

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ *

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation
 - ▶ *

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation
 - ▶ * Lots of good examples

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation
 - ▶ * Lots of good examples
 - ▶ *

The influence of mutualism

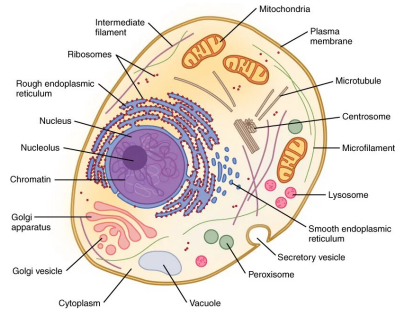
- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Multi-cellularity!

The influence of mutualism

- ▶ Mutualistic and cooperative processes have strongly shaped all of the life that we see
- ▶ Between-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Eukaryotic cells
 - ▶ * Plants
 - ▶ * Pollination and seed dispersal
- ▶ Within-species cooperation
 - ▶ * Lots of good examples
 - ▶ * Multi-cellularity!

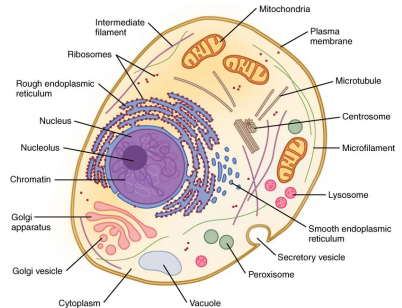
Eukaryotic cells

- The eukaryotic cell descends from at least one ancient mutualism



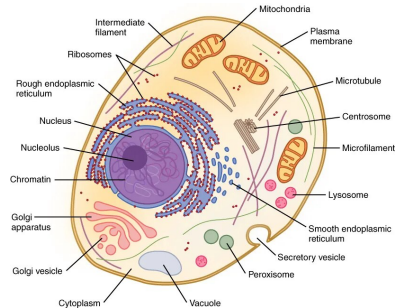
Eukaryotic cells

- ▶ The eukaryotic cell descends from at least one ancient mutualism
- ▶ Land plants descend from a further ancient mutualism



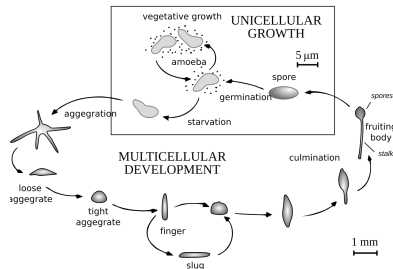
Eukaryotic cells

- ▶ The eukaryotic cell descends from at least one ancient mutualism
- ▶ Land plants descend from a further ancient mutualism



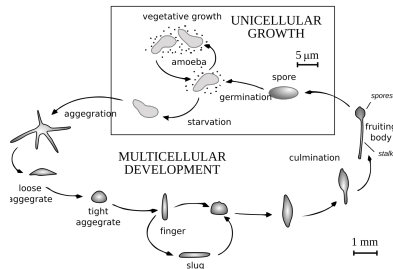
Multi-cellularity

- Multi-cellular organisms descend from ancient communities of co-operators



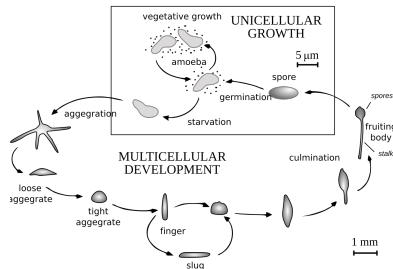
Multi-cellularity

- ▶ Multi-cellular organisms descend from ancient communities of co-operators
- ▶ Evolved separately in plants, animals, fungi and elsewhere

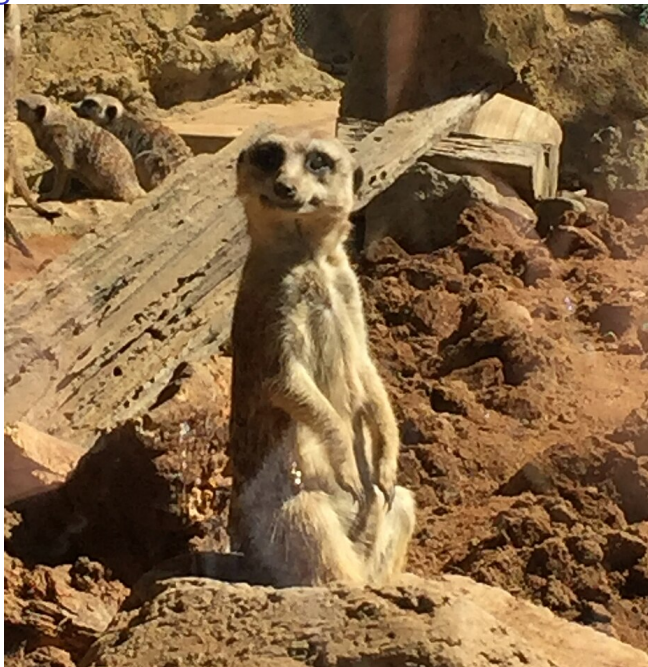


Multi-cellularity

- ▶ Multi-cellular organisms descend from ancient communities of co-operators
- ▶ Evolved separately in plants, animals, fungi and elsewhere



Alarming



Example: Alarming

- ▶ Individuals who spot predators may give alarm calls

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators
 - ▶ Commonly see both within- and between- species

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators
 - ▶ Commonly see both within- and between- species
- ▶ Plants and bacteria also spread alarms

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators
 - ▶ Commonly see both within- and between- species
- ▶ Plants and bacteria also spread alarms
 - ▶ *

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators
 - ▶ Commonly see both within- and between- species
- ▶ Plants and bacteria also spread alarms
 - ▶ * Chemical signalling

Example: Alarming

- ▶ Individuals who spot predators may give alarm calls
- ▶ These benefit other individuals who need to worry about the same predators
 - ▶ Commonly see both within- and between- species
- ▶ Plants and bacteria also spread alarms
 - ▶ * Chemical signalling

Macaque grooming



Beaver grooming



Grooming

- ▶ Individuals help each other stay clean and free of insect parasites

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions
 - ▶ *

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions
 - ▶ * Cleaner fish eat parasites from the skin of larger fish

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions
 - ▶ * Cleaner fish eat parasites from the skin of larger fish
 - ▶ *

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions
 - ▶ * Cleaner fish eat parasites from the skin of larger fish
 - ▶ * Cattle egrets eat parasites from cattle

Grooming

- ▶ Individuals help each other stay clean and free of insect parasites
- ▶ Most commonly occurs within species, but there are exceptions
 - ▶ * Cleaner fish eat parasites from the skin of larger fish
 - ▶ * Cattle egrets eat parasites from cattle

Prisoners dilemma

- ▶ The prisoners' dilemma is a parable for why cooperation can be hard to maintain

B A				
		B stays silent	B testifies	
 A stays silent	 R=1	 R=1	 S=3	 T=0
 A testifies	 T=0	 S=3	 P=2	 P=2

Prisoners dilemma

- ▶ The prisoners' dilemma is a parable for why cooperation can be hard to maintain
- ▶ Two prisoners both know that they will be rewarded for speaking to the police, but will be hurt if the other prisoner speaks to the police

B A \			
		B stays silent	B testifies
	A stays silent	  R=1 R=1	  S=3 T=0
	A testifies	  T=0 S=3	  P=2 P=2

Prisoners dilemma

- ▶ The prisoners' dilemma is a parable for why cooperation can be hard to maintain
- ▶ Two prisoners both know that they will be rewarded for speaking to the police, but will be hurt if the other prisoner speaks to the police
- ▶ What should a selfish prisoner do?

A \ B			
		B stays silent	B testifies
 A stays silent	  R=1 R=1	  S=3 T=0	
 A testifies	  T=0 S=3	  P=2 P=2	

Prisoners dilemma

- ▶ The prisoners' dilemma is a parable for why cooperation can be hard to maintain
- ▶ Two prisoners both know that they will be rewarded for speaking to the police, but will be hurt if the other prisoner speaks to the police
- ▶ What should a selfish prisoner do?
- ▶ Why is this a dilemma?

		B	
		B stays silent	B testifies
A	A stays silent	 R=1 R=1	 S=3 T=0
	A testifies	 T=0 S=3	 P=2 P=2

Prisoners dilemma

- ▶ The prisoners' dilemma is a parable for why cooperation can be hard to maintain
- ▶ Two prisoners both know that they will be rewarded for speaking to the police, but will be hurt if the other prisoner speaks to the police
- ▶ What should a selfish prisoner do?
- ▶ Why is this a dilemma?

		B	
		B stays silent	B testifies
A	A stays silent	 R=1 R=1	 S=3 T=0
	A testifies	 T=0 S=3	 P=2 P=2

Co-ordination and cheating

- The prisoners' dilemma provides a starting point for thinking about two common problems in mutualistic interactions

		B	
A \ B	A	B stays silent	B testifies
	A stays silent	 R=1 R=1	 S=3 T=0
	A testifies	 T=0 S=3	 P=2 P=2

Co-ordination and cheating

- ▶ The prisoners' dilemma provides a starting point for thinking about two common problems in mutualistic interactions
 - ▶ Co-ordination: how does cooperation start?

		B	
A \ B	A stays silent	B stays silent	B testifies
A stays silent	 A stays silent	 R=1 R=1	 S=3 T=0
A testifies	 A testifies	 T=0 S=3	 P=2 P=2

Co-ordination and cheating

- ▶ The prisoners' dilemma provides a starting point for thinking about two common problems in mutualistic interactions
 - ▶ Co-ordination: how does cooperation start?
 - ▶ Cheating: When can cooperators be taken advantage of?

		B	
		A stays silent	A testifies
A	B stays silent	 R=1 R=1	 S=3 T=0
	B testifies	 T=0 S=3	 P=2 P=2

Co-ordination and cheating

- ▶ The prisoners' dilemma provides a starting point for thinking about two common problems in mutualistic interactions
 - ▶ Co-ordination: how does cooperation start?
 - ▶ Cheating: When can cooperators be taken advantage of?

		B	
		B stays silent	B testifies
A	A stays silent	 R=1 R=1	 S=3 T=0
A	A testifies	 T=0 S=3	 P=2 P=2

Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances

Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances
- ▶ These “producers” are not killed by the colicin, but “sensitive” relatives are, giving producers an advantage

Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances
- ▶ These “producers” are not killed by the colicin, but “sensitive” relatives are, giving producers an advantage
- ▶ What does this have to do with co-operation?

Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances
- ▶ These “producers” are not killed by the colicin, but “sensitive” relatives are, giving producers an advantage
- ▶ What does this have to do with co-operation?



Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances
- ▶ These “producers” are not killed by the colicin, but “sensitive” relatives are, giving producers an advantage
- ▶ What does this have to do with co-operation?
 - ▶ * When bacteria produce colicin, they are helping any “resistant” bacteria (not killed by colicin) in their area

Colicin

- ▶ Many bacteria have the ability to produce a toxin called “colicin”, which they release into the environment under certain circumstances
- ▶ These “producers” are not killed by the colicin, but “sensitive” relatives are, giving producers an advantage
- ▶ What does this have to do with co-operation?
 - ▶ * When bacteria produce colicin, they are helping any “resistant” bacteria (not killed by colicin) in their area

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ *

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ *

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)
 - ▶ *

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)
 - ▶ * producers might invade again!

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)
 - ▶ * producers might invade again!
 - ▶ *

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)
 - ▶ * producers might invade again!
 - ▶ * rock-paper-scissors dynamics!

Colicin dynamics

- ▶ Assume that producing colicin has a cost, and that resisting colicin has a separate cost
- ▶ A population of sensitive bacteria is invaded by a population of producers
 - ▶ Killed off by toxin
- ▶ What might happen next?
 - ▶ * resistant bacteria might successfully invade the producers (they are more efficient)
 - ▶ * sensitive bacteria might successfully invade the resisters (same reason)
 - ▶ * producers might invade again!
 - ▶ * rock-paper-scissors dynamics!

Outline

Introduction

Cooperation within species

Mutualism between species

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ *

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ *

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ * There are less likely to be strong relative advantages (things I can do that you can't)

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ * There are less likely to be strong relative advantages (things I can do that you can't)
- ▶ What are some possible explanations?

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ * There are less likely to be strong relative advantages (things I can do that you can't)
- ▶ What are some possible explanations?
 - ▶ *

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ * There are less likely to be strong relative advantages (things I can do that you can't)
- ▶ What are some possible explanations?
 - ▶ *

Obstacles

- ▶ How can helping evolve if there is a cost to the helper?
- ▶ This is particularly hard to explain within species:
 - ▶ * Individuals of the same species are my strongest competitors
 - ▶ * There are less likely to be strong relative advantages (things I can do that you can't)
- ▶ What are some possible explanations?
 - ▶ *

Absence of cost

- ▶ Behaviour that helps others may also help the focal individual



Absence of cost

- ▶ Behaviour that helps others may also help the focal individual
- ▶ Alarm calls may have evolved to communicate with *predators*



Absence of cost

- ▶ Behaviour that helps others may also help the focal individual
- ▶ Alarm calls may have evolved to communicate with *predators*



Absence of cost

- ▶ Behaviour that helps others may also help the focal individual
- ▶ Alarm calls may have evolved to communicate with *predators*
 - ▶ * I've seen you, don't bother me



Absence of cost

- ▶ Behaviour that helps others may also help the focal individual
- ▶ Alarm calls may have evolved to communicate with *predators*
 - ▶ * I've seen you, don't bother me



Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?



Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ *

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?
 - ▶ *

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?
 - ▶ * Not really, cheaters could rely on others to spot the predators

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?
 - ▶ * Not really, cheaters could rely on others to spot the predators
 - ▶ *

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?
 - ▶ * Not really, cheaters could rely on others to spot the predators
 - ▶ * If all of the other zebra are watching for predators, I can just watch the other zebra

Co-ordination and cheating

- ▶ Does absence of cost solve the co-ordination problem?
 - ▶ * Yes, I call for my own good and you for yours
 - ▶ * We can learn to respond to each others' calls
- ▶ Does absence of cost solve the cheating problem?
 - ▶ * Not really, cheaters could rely on others to spot the predators
 - ▶ * If all of the other zebra are watching for predators, I can just watch the other zebra

Group selection

- ▶ Costly behaviours might be selected for at the level of groups

Group selection

- ▶ Costly behaviours might be selected for at the level of groups
- ▶ Groups where cheaters arise might eventually be out-competed by groups where they don't

Group selection

- ▶ Costly behaviours might be selected for at the level of groups
- ▶ Groups where cheaters arise might eventually be out-competed by groups where they don't
- ▶ Depends on the effective level at which natural selection is acting

Group selection

- ▶ Costly behaviours might be selected for at the level of groups
- ▶ Groups where cheaters arise might eventually be out-competed by groups where they don't
- ▶ Depends on the effective level at which natural selection is acting

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness
 - ▶ Fitness is eventually achieved through offspring

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness
 - ▶ Fitness is eventually achieved through offspring
 - ▶ But full siblings are just as related to me as my offspring are!

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness
 - ▶ Fitness is eventually achieved through offspring
 - ▶ But full siblings are just as related to me as my offspring are!
 - ▶ Evolution selects to favor individuals reproducing their genes

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness
 - ▶ Fitness is eventually achieved through offspring
 - ▶ But full siblings are just as related to me as my offspring are!
 - ▶ Evolution selects to favor individuals reproducing their genes
 - ▶ not necessarily having their own offspring

Kin selection

- ▶ Costly behaviours can be selected if they benefit *relatives* (kin)
 - ▶ Inclusive fitness
 - ▶ Fitness is eventually achieved through offspring
 - ▶ But full siblings are just as related to me as my offspring are!
 - ▶ Evolution selects to favor individuals reproducing their genes
 - ▶ not necessarily having their own offspring

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:
 - ▶ Benefit to relatives multiplied by relatedness exceeds cost to individual

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:
 - ▶ Benefit to relatives multiplied by relatedness exceeds cost to individual
- ▶ You should see this as a loose guide

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:
 - ▶ Benefit to relatives multiplied by relatedness exceeds cost to individual
- ▶ You should see this as a loose guide
- ▶ *I would gladly lay down my life for my two siblings*

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:
 - ▶ Benefit to relatives multiplied by relatedness exceeds cost to individual
- ▶ You should see this as a loose guide
- ▶ *I would gladly lay down my life for my two siblings*
 - ▶ *or for eight cousins*

Hamilton's rule

- ▶ Costly sharing is good if the relative benefit outweighs the cost
- ▶ A costly act is favored by natural selection if $rb > c$:
 - ▶ Benefit to relatives multiplied by relatedness exceeds cost to individual
- ▶ You should see this as a loose guide
- ▶ *I would gladly lay down my life for my two siblings*
 - ▶ *or for eight cousins*

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of "relatedness"?

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population
- ▶ Why do we assume we're always competing with a relevant population?

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population
- ▶ Why do we assume we're always competing with a relevant population?
 - ▶ *

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population
- ▶ Why do we assume we're always competing with a relevant population?
 - ▶ * Population regulation, the “third law” of population dynamics

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population
- ▶ Why do we assume we're always competing with a relevant population?
 - ▶ * Population regulation, the “third law” of population dynamics
- ▶ It is hard to disentangle group selection from kin selection

Limitations

- ▶ Hamilton's rule suggests that pure altruism (paying a cost for others) can be selected for
- ▶ It is hard to apply Hamilton's rule quantitatively
- ▶ Benefits and costs are hard to estimate
- ▶ What is the effective value of “relatedness”?
 - ▶ How much *more* related is the altruist to the beneficiary than to the average individual in the relevant population
- ▶ Why do we assume we're always competing with a relevant population?
 - ▶ * Population regulation, the “third law” of population dynamics
- ▶ It is hard to disentangle group selection from kin selection

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?
 - ▶ *

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?
 - ▶ * Avoidance of inbreeding

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?
 - ▶ * Avoidance of inbreeding
 - ▶ *

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?
 - ▶ * Avoidance of inbreeding
 - ▶ * if recognition exists, it can be used for other purposes

Kin Recognition

- ▶ Kin-based selection suggests organisms should help kin more than non-kin
- ▶ How do organisms recognize kin?
 - ▶ Siblings, parents and offspring from context
 - ▶ Chemical cues
- ▶ Is there another reason many organisms have evolved to recognize kin?
 - ▶ * Avoidance of inbreeding
 - ▶ * if recognition exists, it can be used for other purposes

Example: Plant recognition experiments

- ▶ Plants grown next to kin compete less aggressively



Example: Plant recognition experiments

- ▶ Plants grown next to kin compete less aggressively
- ▶ Work from Susan Dudley, McMaster Biology



Example: Plant recognition experiments

- ▶ Plants grown next to kin compete less aggressively
- ▶ Work from Susan Dudley, McMaster Biology



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:
 - ▶ * Building biofilms



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:
 - ▶ * Building biofilms
 - ▶ *



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:
 - ▶ * Building biofilms
 - ▶ * Making antibiotics



Kin viscosity

- ▶ Kin viscosity is the idea that any same-species individuals you encounter are likely to be related
- ▶ Activities that benefit the group are likely to favor your fitness
- ▶ Many micro-organisms have behaviours that seem to make use of this:
 - ▶ * Building biofilms
 - ▶ * Making antibiotics



Gene-level view

- ▶ We've thought about the group level and the kin-fitness level

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ *

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ * A trait that others can recognize, which signals the presence of a co-operator gene

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ * A trait that others can recognize, which signals the presence of a co-operator gene
 - ▶ What about cheating?

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ * A trait that others can recognize, which signals the presence of a co-operator gene
 - ▶ What about cheating?
 - ▶ *

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ * A trait that others can recognize, which signals the presence of a co-operator gene
 - ▶ What about cheating?
 - ▶ * A gene that makes co-operators think you have the co-operator gene

Gene-level view

- ▶ We've thought about the group level and the kin-fitness level
- ▶ It might be that *genes* are the right level, though
- ▶ The real question about whether a gene for helping others will spread is whether those others have that same gene
 - ▶ How would you know?
 - ▶ Does this select for sets of linked traits?
 - ▶ * A trait that others can recognize, which signals the presence of a co-operator gene
 - ▶ What about cheating?
 - ▶ * A gene that makes co-operators think you have the co-operator gene

Reciprocity

- ▶ Another way helping could be favored selectively is organisms help each other

Reciprocity

- ▶ Another way helping could be favored selectively is organisms help each other
 - ▶ Two individuals could explore each others' willingness to help out

Reciprocity

- ▶ Another way helping could be favored selectively is organisms help each other
 - ▶ Two individuals could explore each others' willingness to help out
 - ▶ If I help you, you help me, and we both benefit

Reciprocity

- ▶ Another way helping could be favored selectively is organisms help each other
 - ▶ Two individuals could explore each others' willingness to help out
 - ▶ If I help you, you help me, and we both benefit
- ▶ Social organisms may have many, many opportunities to interact

Reciprocity

- ▶ Another way helping could be favored selectively is organisms help each other
 - ▶ Two individuals could explore each others' willingness to help out
 - ▶ If I help you, you help me, and we both benefit
- ▶ Social organisms may have many, many opportunities to interact

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then
 - ▶ If I assume you have the same logic about the 10th time, then the same logic applies to the 9th time as well ...

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then
 - ▶ If I assume you have the same logic about the 10th time, then the same logic applies to the 9th time as well ...
- ▶ In experimental competitions, however, “virtual organisms” with intermediate strategies do best.

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then
 - ▶ If I assume you have the same logic about the 10th time, then the same logic applies to the 9th time as well ...
- ▶ In experimental competitions, however, “virtual organisms” with intermediate strategies do best.
 - ▶ This may reflect biology: evolution is not perfect, it is just doing the best that it can

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then
 - ▶ If I assume you have the same logic about the 10th time, then the same logic applies to the 9th time as well ...
- ▶ In experimental competitions, however, “virtual organisms” with intermediate strategies do best.
 - ▶ This may reflect biology: evolution is not perfect, it is just doing the best that it can
- ▶ Additionally, real organisms don't know exactly how many times they will interact

Example: Prisoners' dilemma

- ▶ According to simple logic, if you and I are going to play the Prisoners dilemma 10 times, my best strategy is to be selfish
 - ▶ Obviously, whatever you do the 10th time, I should be selfish then
 - ▶ If I assume you have the same logic about the 10th time, then the same logic applies to the 9th time as well ...
- ▶ In experimental competitions, however, “virtual organisms” with intermediate strategies do best.
 - ▶ This may reflect biology: evolution is not perfect, it is just doing the best that it can
- ▶ Additionally, real organisms don't know exactly how many times they will interact



Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming

▶ *

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:
 - ▶ A form of cheating – I enjoy the benefits of help, but don't myself help

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:
 - ▶ A form of cheating – I enjoy the benefits of help, but don't myself help
 - ▶ If free riders have higher fitness, what can happen?

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:
 - ▶ A form of cheating – I enjoy the benefits of help, but don't myself help
 - ▶ If free riders have higher fitness, what can happen?
 - ▶ *

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:
 - ▶ A form of cheating – I enjoy the benefits of help, but don't myself help
 - ▶ If free riders have higher fitness, what can happen?
 - ▶ * It's complicated

Public goods

- ▶ Does the idea of reciprocity apply equally to all forms of helping?
 - ▶ e.g., grooming vs. alarming
 - ▶ * Alarming is shared more generally
- ▶ Widely shared **public goods** are subject to **free riding**:
 - ▶ A form of cheating – I enjoy the benefits of help, but don't myself help
 - ▶ If free riders have higher fitness, what can happen?
 - ▶ * It's complicated

Enforcement

- ▶ Animals can take steps to *punish* free riders

Enforcement

- ▶ Animals can take steps to *punish* free riders
 - ▶ Meerkats that don't watch may be driven out

Enforcement

- ▶ Animals can take steps to *punish* free riders
 - ▶ Meerkats that don't watch may be driven out
 - ▶ lionesses that don't defend territory may be attacked

Enforcement

- ▶ Animals can take steps to *punish* free riders
 - ▶ Meerkats that don't watch may be driven out
 - ▶ lionesses that don't defend territory may be attacked
 - ▶ ... evidence is limited

Enforcement

- ▶ Animals can take steps to *punish* free riders
 - ▶ Meerkats that don't watch may be driven out
 - ▶ lionesses that don't defend territory may be attacked
 - ▶ ... evidence is limited
- ▶ Social animals that cooperate may also be rewarded with higher status

Enforcement

- ▶ Animals can take steps to *punish* free riders
 - ▶ Meerkats that don't watch may be driven out
 - ▶ lionesses that don't defend territory may be attacked
 - ▶ ... evidence is limited
- ▶ Social animals that cooperate may also be rewarded with higher status

Mobbing



Balance

- ▶ If a behaviour helps individuals (like calling to prevent predator harassment), then free riding is unlikely to eliminate it

Balance

- ▶ If a behaviour helps individuals (like calling to prevent predator harassment), then free riding is unlikely to eliminate it
- ▶ Cheating may be balanced, or eliminated, by kin selection

Balance

- ▶ If a behaviour helps individuals (like calling to prevent predator harassment), then free riding is unlikely to eliminate it
- ▶ Cheating may be balanced, or eliminated, by kin selection

Fitness capture

- ▶ If evolutionary changes make it harder to cheat, then cooperative behaviours may evolve more efficiently

Fitness capture

- ▶ If evolutionary changes make it harder to cheat, then cooperative behaviours may evolve more efficiently
- ▶ Many colicin-expressing bacteria are stable; they have configurations that make it hard for them to maintain resistance without at least sometimes producing colicin

Fitness capture

- ▶ If evolutionary changes make it harder to cheat, then cooperative behaviours may evolve more efficiently
- ▶ Many colicin-expressing bacteria are stable; they have configurations that make it hard for them to maintain resistance without at least sometimes producing colicin

Social insects

- ▶ Some insects are highly social

Social insects

- ▶ Some insects are highly social
- ▶ Colonies of millions of workers, but all (or virtually all) fitness is via the queen and the drones

Social insects

- ▶ Some insects are highly social
- ▶ Colonies of millions of workers, but all (or virtually all) fitness is via the queen and the drones
- ▶ Very complex adaptations keep this stable: allowing other very complex adaptations for efficient colonies

Social insects

- ▶ Some insects are highly social
- ▶ Colonies of millions of workers, but all (or virtually all) fitness is via the queen and the drones
- ▶ Very complex adaptations keep this stable: allowing other very complex adaptations for efficient colonies

Termite mound



Cellular slime molds

► Are crazy!

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi
- ▶ Individual “amoebas” join up for dispersal and reproduction

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi
- ▶ Individual “amoebas” join up for dispersal and reproduction
 - ▶ Amoebas that wind up in the stalk of a fruiting body are sacrificing their own reproduction

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi
- ▶ Individual “amoebas” join up for dispersal and reproduction
 - ▶ Amoebas that wind up in the stalk of a fruiting body are sacrificing their own reproduction
- ▶ A wide variety of mechanisms to make cheating difficult

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi
- ▶ Individual “amoebas” join up for dispersal and reproduction
 - ▶ Amoebas that wind up in the stalk of a fruiting body are sacrificing their own reproduction
- ▶ A wide variety of mechanisms to make cheating difficult
 - ▶ Combined with kin selection

Cellular slime molds

- ▶ Are crazy!
- ▶ Sometimes act like micro-organisms; sometimes like multi-cellular animals, and sometimes like plants or fungi
- ▶ Individual “amoebas” join up for dispersal and reproduction
 - ▶ Amoebas that wind up in the stalk of a fruiting body are sacrificing their own reproduction
- ▶ A wide variety of mechanisms to make cheating difficult
 - ▶ Combined with kin selection

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends
 - ▶ They only have kin-based fitness

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends
 - ▶ They only have kin-based fitness
- ▶ Disease biology provides important insights into the process

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends
 - ▶ They only have kin-based fitness
- ▶ Disease biology provides important insights into the process
 - ▶ Infected or cancerous cells may be targeted and killed by other cells

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends
 - ▶ They only have kin-based fitness
- ▶ Disease biology provides important insights into the process
 - ▶ Infected or cancerous cells may be targeted and killed by other cells
 - ▶ Or they may commit suicide!

Modern multi-cellular organisms

- ▶ We've been through hundreds of millions of years of evolution since our fitness became linked
 - ▶ Kidney cells or muscle cells are absolute, evolutionary dead ends
 - ▶ They only have kin-based fitness
- ▶ Disease biology provides important insights into the process
 - ▶ Infected or cancerous cells may be targeted and killed by other cells
 - ▶ Or they may commit suicide!

Outline

Introduction

Cooperation within species

Mutualism between species

Mutualism between species

- ▶ May evolve more easily than within a species

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ *

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ * Fungi and algae make lichens

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ * Fungi and algae make lichens
 - ▶ *

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ * Fungi and algae make lichens
 - ▶ * Plants can photosynthesize, but use animals to move themselves or their pollen around

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ * Fungi and algae make lichens
 - ▶ * Plants can photosynthesize, but use animals to move themselves or their pollen around
- ▶ Still need to worry about co-ordination and cheating

Mutualism between species

- ▶ May evolve more easily than within a species
 - ▶ Often not competing directly for resources
 - ▶ More likely to have complementary abilities:
 - ▶ * Fungi and algae make lichens
 - ▶ * Plants can photosynthesize, but use animals to move themselves or their pollen around
- ▶ Still need to worry about co-ordination and cheating

One-way help

- ▶ Birds nesting in trees

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ *

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ * Since they get no benefit, there is strong selection against paying a cost

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ * Since they get no benefit, there is strong selection against paying a cost
- ▶ Cheating is not possible in these cases

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ * Since they get no benefit, there is strong selection against paying a cost
- ▶ Cheating is not possible in these cases
 - ▶ *

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ * Since they get no benefit, there is strong selection against paying a cost
- ▶ Cheating is not possible in these cases
 - ▶ * **Beneficiary** is distinct from the helper

One-way help

- ▶ Birds nesting in trees
- ▶ Poop provides food for dung beetles
- ▶ Typically no cost to helper
 - ▶ * Since they get no benefit, there is strong selection against paying a cost
- ▶ Cheating is not possible in these cases
 - ▶ * **Beneficiary** is distinct from the helper

Symmetric help

- ▶ Things that are poisonous or otherwise hard to eat benefit from looking obvious, so that they don't get attacked

Symmetric help

- ▶ Things that are poisonous or otherwise hard to eat benefit from looking obvious, so that they don't get attacked
 - ▶ They also benefit from looking like *each other*

Symmetric help

- ▶ Things that are poisonous or otherwise hard to eat benefit from looking obvious, so that they don't get attacked
 - ▶ They also benefit from looking like *each other*
 - ▶ This is called **Mullerian mimicry**

Symmetric help

- ▶ Things that are poisonous or otherwise hard to eat benefit from looking obvious, so that they don't get attacked
 - ▶ They also benefit from looking like *each other*
 - ▶ This is called **Mullerian mimicry**
- ▶ Co-ordination is not a problem, since each species wants to be recognized

Symmetric help

- ▶ Things that are poisonous or otherwise hard to eat benefit from looking obvious, so that they don't get attacked
 - ▶ They also benefit from looking like *each other*
 - ▶ This is called **Mullerian mimicry**
- ▶ Co-ordination is not a problem, since each species wants to be recognized

Mullerian mimicry



Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ *

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ * Free riding

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ * Free riding
- ▶ Batesian mimics tend to be rare: why?

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ * Free riding
- ▶ Batesian mimics tend to be rare: why?
 - ▶ *

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ * Free riding
- ▶ Batesian mimics tend to be rare: why?
 - ▶ * Predators will learn not to be afraid if harmless species is common

Batesian mimicry

- ▶ Similar organisms that are *not* dangerous also benefit from looking like the nasty ones
- ▶ This is an example of
 - ▶ * Free riding
- ▶ Batesian mimics tend to be rare: why?
 - ▶ * Predators will learn not to be afraid if harmless species is common

Batesian mimicry

**Batesian
Mimic
(parasitic benefit)**



**Deceptive signal
Hoverfly harmless**



**Müllerian
Mimics
(mutual benefit)**



**Honest signal
Wasp can sting**



**Honest signal
Bee can sting**

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours
 - ▶ Again, co-ordination is not a problem

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours
 - ▶ Again, co-ordination is not a problem
 - ▶ What about free riding?

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours
 - ▶ Again, co-ordination is not a problem
 - ▶ What about free riding?
 - ▶ *

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours
 - ▶ Again, co-ordination is not a problem
 - ▶ What about free riding?
 - ▶ * Yes. Species or individuals could rely on others to be alert

Mixed flocks and schools

- ▶ Different species of birds or fish may associate with each other and respond to each others' alarm behaviours
 - ▶ Again, co-ordination is not a problem
 - ▶ What about free riding?
 - ▶ * Yes. Species or individuals could rely on others to be alert

Relative advantage

- ▶ Different organisms with different abilities can work together

Relative advantage

- ▶ Different organisms with different abilities can work together



Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ *

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ *

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders
 - ▶ *

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders
 - ▶ * Plants and seed disperses

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders
 - ▶ * Plants and seed disperses
 - ▶ *

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders
 - ▶ * Plants and seed disperses
 - ▶ * So many others!

Relative advantage

- ▶ Different organisms with different abilities can work together
 - ▶ * Eukaryotes and their captured organelles
 - ▶ * Mitochondria, chloroplasts, apicoplasts
 - ▶ * Plants and pollen spreaders
 - ▶ * Plants and seed disperses
 - ▶ * So many others!

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating
- ▶ An organism may take advantage of provided resources without providing a service

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating
- ▶ An organism may take advantage of provided resources without providing a service
 - ▶ Hummingbirds may drink nectar without spreading pollen

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating
- ▶ An organism may take advantage of provided resources without providing a service
 - ▶ Hummingbirds may drink nectar without spreading pollen
 - ▶ Or orchids might attract insects without providing useful food resources

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating
- ▶ An organism may take advantage of provided resources without providing a service
 - ▶ Hummingbirds may drink nectar without spreading pollen
 - ▶ Or orchids might attract insects without providing useful food resources
 - ▶ Sometimes by mimicking female insects

Trading

- ▶ Trades based on relative advantage very often have the possibility of cheating
- ▶ An organism may take advantage of provided resources without providing a service
 - ▶ Hummingbirds may drink nectar without spreading pollen
 - ▶ Or orchids might attract insects without providing useful food resources
 - ▶ Sometimes by mimicking female insects

Hummingbird flower



Example: ant-plant mutualisms

- ▶ Some trees provide resources to ants who live on them and defend them from other things that want to eat them

Example: ant-plant mutualisms

- ▶ Some trees provide resources to ants who live on them and defend them from other things that want to eat them
- ▶ Some of these are highly developed, and **obligatory** for both species

Example: ant-plant mutualisms

- ▶ Some trees provide resources to ants who live on them and defend them from other things that want to eat them
- ▶ Some of these are highly developed, and **obligatory** for both species
 - ▶ Ants live nowhere else

Example: ant-plant mutualisms

- ▶ Some trees provide resources to ants who live on them and defend them from other things that want to eat them
- ▶ Some of these are highly developed, and **obligatory** for both species
 - ▶ Ants live nowhere else
 - ▶ Trees do not thrive without ants

Example: ant-plant mutualisms

- ▶ Some trees provide resources to ants who live on them and defend them from other things that want to eat them
- ▶ Some of these are highly developed, and **obligatory** for both species
 - ▶ Ants live nowhere else
 - ▶ Trees do not thrive without ants

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:
 - ▶ special thorns for them to live in

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:
 - ▶ special thorns for them to live in
 - ▶ special nectaries and protein sources

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:
 - ▶ special thorns for them to live in
 - ▶ special nectaries and protein sources
- ▶ They protect themselves from exploitation by not producing these things unless the ants signal them

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:
 - ▶ special thorns for them to live in
 - ▶ special nectaries and protein sources
- ▶ They protect themselves from exploitation by not producing these things unless the ants signal them
- ▶ There are several species of cheater ants anyway; these mimic the mutualistic ants

Ant-plant mutualisms

- ▶ Trees have a wide variety of resources just for the ants:
 - ▶ special thorns for them to live in
 - ▶ special nectaries and protein sources
- ▶ They protect themselves from exploitation by not producing these things unless the ants signal them
- ▶ There are several species of cheater ants anyway; these mimic the mutualistic ants

Acacia ants



Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely
- ▶ Fruits with thick skins

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely
 - ▶ Fruits with thick skins
 - ▶ *

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely
 - ▶ Fruits with thick skins
 - ▶ * Less likely to be penetrated by small things that won't spread them

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely
 - ▶ Fruits with thick skins
 - ▶ * Less likely to be penetrated by small things that won't spread them
- ▶ *Avocadoes may be fatty to discourage consumption by normal-sized things*

Evolution and co-evolution

- ▶ Many species have elaborate defenses against cheaters
 - ▶ Bacteria have to pass a large number of tests to enter and survive in squid light organs; squids expel them frequently anyway
 - ▶ Plants punish or reward bacterial nodules depending on how much nitrogen they fix
 - ▶ If nitrogen is abundant, they may shut down the nodules entirely
 - ▶ Fruits with thick skins
 - ▶ * Less likely to be penetrated by small things that won't spread them
 - ▶ *Avocadoes may be fatty to discourage consumption by normal-sized things*

Balance

- ▶ Many mutualisms survive in balance with cheaters

Balance

- ▶ Many mutualisms survive in balance with cheaters
- ▶ Cheaters may do well only when in low abundance

Balance

- ▶ Many mutualisms survive in balance with cheaters
- ▶ Cheaters may do well only when in low abundance
 - ▶ Batesian mimics

Balance

- ▶ Many mutualisms survive in balance with cheaters
- ▶ Cheaters may do well only when in low abundance
 - ▶ Batesian mimics
 - ▶ “Pirates” of ant-plant interactions

Balance

- ▶ Many mutualisms survive in balance with cheaters
- ▶ Cheaters may do well only when in low abundance
 - ▶ Batesian mimics
 - ▶ “Pirates” of ant-plant interactions
 - ▶ Need to make a profit quickly and move on

Balance

- ▶ Many mutualisms survive in balance with cheaters
- ▶ Cheaters may do well only when in low abundance
 - ▶ Batesian mimics
 - ▶ “Pirates” of ant-plant interactions
 - ▶ Need to make a profit quickly and move on

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists
 - ▶ ants and some acacia trees

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists
 - ▶ ants and some acacia trees
 - ▶ root fungi and plants

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists
 - ▶ ants and some acacia trees
 - ▶ root fungi and plants
- ▶ Others might have very weak relationships

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists
 - ▶ ants and some acacia trees
 - ▶ root fungi and plants
- ▶ Others might have very weak relationships
 - ▶ Vertebrates eating fruit from a particular plant

Linkage

- ▶ Similar to within-species cooperation, there can be a lot of conflict when fitness is not well linked, and more efficient evolution when species become linked more closely
- ▶ Some species can *only* survive with particular mutualists
 - ▶ ants and some acacia trees
 - ▶ root fungi and plants
- ▶ Others might have very weak relationships
 - ▶ Vertebrates eating fruit from a particular plant



Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!

Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!
- ▶ They are completely dependent on highly specialized fungi that they raise

Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!
- ▶ They are completely dependent on highly specialized fungi that they raise
 - ▶ These fungi have no ability to survive without ants

Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!
- ▶ They are completely dependent on highly specialized fungi that they raise
 - ▶ These fungi have no ability to survive without ants
- ▶ The association is highly evolved

Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!
- ▶ They are completely dependent on highly specialized fungi that they raise
 - ▶ These fungi have no ability to survive without ants
- ▶ The association is highly evolved
 - ▶ The ants culture bacteria that attack cheating fungi, but not their own fungi

Example: leaf-cutter ants

- ▶ Leaf cutter ants have no ability to digest leaves!
- ▶ They are completely dependent on highly specialized fungi that they raise
 - ▶ These fungi have no ability to survive without ants
- ▶ The association is highly evolved
 - ▶ The ants culture bacteria that attack cheating fungi, but not their own fungi



Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent



Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!
 - ▶ *

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!
 - ▶ * Suppress production of offspring which don't include them

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!
 - ▶ * Suppress production of offspring which don't include them
 - ▶ *

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!
 - ▶ * Suppress production of offspring which don't include them
 - ▶ * For example, by favoring a female-biased sex ratio in less-complex eukaryotes

Example: Mitochondria

- ▶ We don't think of mitochondria as having independent fitness, but there is almost always some potential for conflict in biology
- ▶ Most eukaryotes have mitochondria inherited from a single parent
 - ▶ * To reduce mitochondrial competition that would be bad for the eukaryote
- ▶ Sometimes mitochondria are able to cause conflict with the eukaryote anyway!
 - ▶ * Suppress production of offspring which don't include them
 - ▶ * For example, by favoring a female-biased sex ratio in less-complex eukaryotes

Conclusions

- ▶ Positive interactions are pervasive

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict
 - ▶ endo-symbiosis, multi-cellularity

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict
 - ▶ endo-symbiosis, multi-cellularity
- ▶ There are also examples where conflict emerges anyway!

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict
 - ▶ endo-symbiosis, multi-cellularity
- ▶ There are also examples where conflict emerges anyway!
 - ▶ *

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict
 - ▶ endo-symbiosis, multi-cellularity
- ▶ There are also examples where conflict emerges anyway!
 - ▶ * cancer and mitochondrial sex selection

Conclusions

- ▶ Positive interactions are pervasive
 - ▶ And a major driver of evolution
- ▶ Opportunities for conflict arise often
 - ▶ Much of life is a balance between co-operation and conflict
- ▶ Sometimes, evolutionary changes can minimize conflict
 - ▶ endo-symbiosis, multi-cellularity
- ▶ There are also examples where conflict emerges anyway!
 - ▶ * cancer and mitochondrial sex selection